# Week 3 Lesson 2 — Sigmoid function

31st October 2026

## 1 The Sigmoid Function Clearly Explained

Notebook done to better understand The Sigmoid Function Clearly Explained

The sigmoid function is defined as

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

It is denoted by the Greek letter *sigma*.

Reference: `numpy.exp`

Calculate the exponential of all elements in the input array.

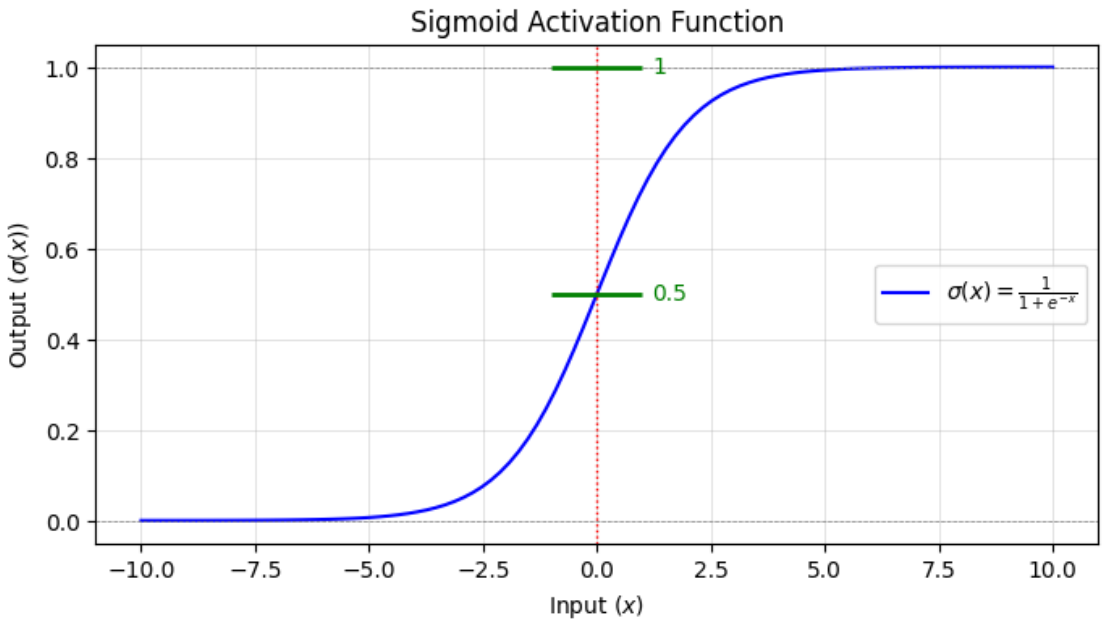
[66]:
```
import matplotlib.pyplot as plt
import numpy as np


# Define the sigmoid function
def sigmoid(x):
  return 1 / (1 + np.exp(-x))


# Generate x values from -10 to 10
x = np.linspace(-10, 10, 100)
y = sigmoid(x)

# Plot the S-shaped curve
plt.figure(figsize=(8, 4))
plt.plot(x, y, label=r"$\sigma(x) = \frac{1}{1 + e^{-x}}$", color="blue")
plt.axhline(0, color="gray", linestyle="--", linewidth=0.5)
plt.axhline(1, color="gray", linestyle="--", linewidth=0.5)
plt.axvline(0, color="red", linestyle=":", linewidth=1)
plt.title("Sigmoid Activation Function")
plt.xlabel("Input ($x$)")
plt.ylabel(r"Output ($\sigma(x)$)")
plt.grid(True)
plt.legend()
plt.hlines(y=0.5, xmin=-1, xmax=1, color="green", linewidth=2)
```

```
plt.text(1.2, 0.5, "0.5", color="green", va="center")
plt.hlines(y=1, xmin=-1, xmax=1, color="green", linewidth=2)
plt.text(1.2, 1, "1", color="green", va="center")
plt.show()
```




Exponential functions are functions of the form $f(x) = a^x$, where $a$ is a positive constant. The sigmoid function is a special case of an exponential function.

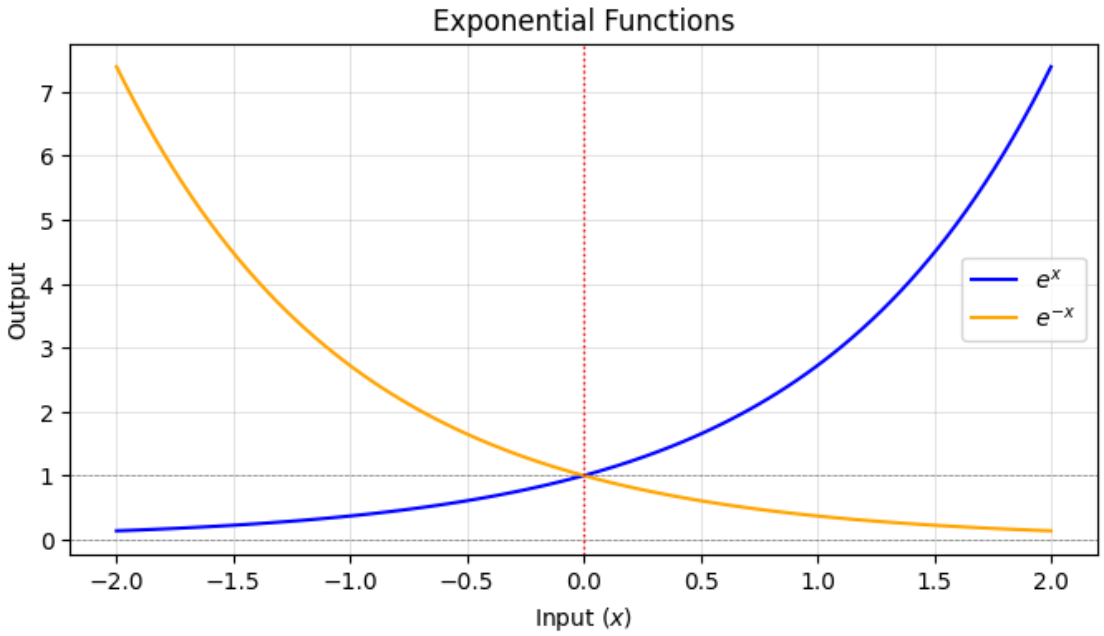
Let us plot:

$$e^{-x} = \frac{1}{e^x}$$

[67]:
```
x = np.linspace(-2, 2, 100)
y_exp = np.exp(x)
y_exp_negative = np.exp(-x)

# Plot the exponential functions
plt.figure(figsize=(8, 4))
plt.plot(x, y_exp, label=r"$e^x$", color="blue")
plt.plot(x, y_exp_negative, label=r"$e^{-x}$", color="orange")
plt.axhline(0, color="gray", linestyle="--", linewidth=0.5)
plt.axhline(1, color="gray", linestyle="--", linewidth=0.5)
plt.axvline(0, color="red", linestyle=":", linewidth=1)
plt.title("Exponential Functions")
plt.xlabel("Input ($x$)")
plt.ylabel("Output")
```

```
plt.grid(True)
plt.legend()
plt.show()
```



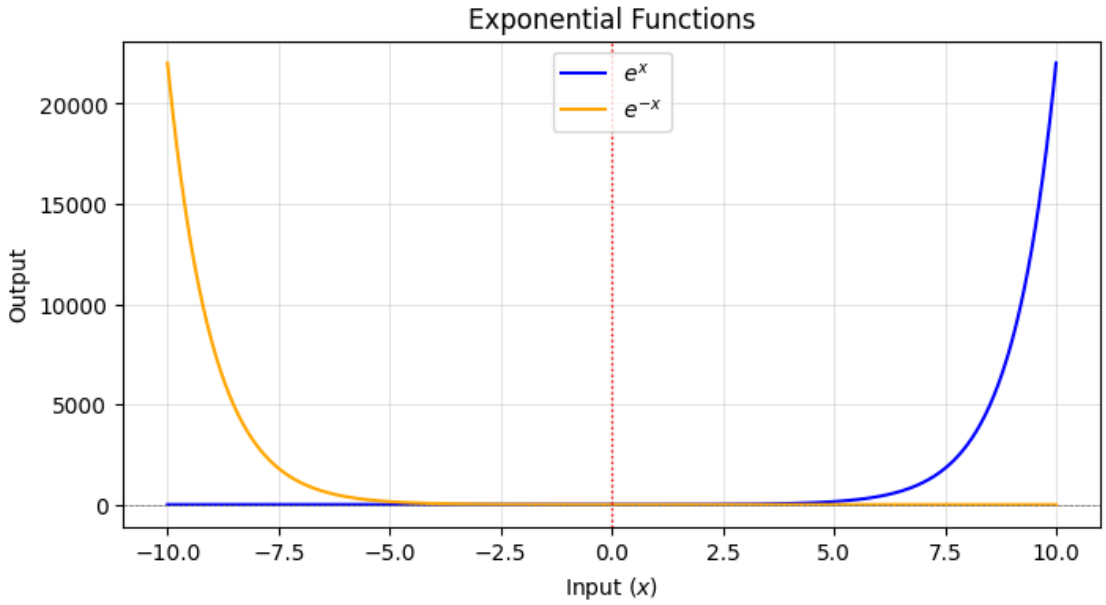

It can be seen how it grows towards infinity, and converges to 1 on x=0:

$$\lim_{x \to -\infty} e^{-x} = +\infty$$
$$\lim_{x \to 0} e^{-x} = 1$$
$$\lim_{x \to \infty} e^{-x} = 0$$

[68]:
```
x = np.linspace(-10, 10, 1001)
y_exp = np.exp(x)
y_exp_negative = np.exp(-x)

# Plot the exponential functions
plt.figure(figsize=(8, 4))
plt.plot(x, y_exp, label=r"$e^x$", color="blue")
plt.plot(x, y_exp_negative, label=r"$e^{-x}$", color="orange")
plt.axhline(0, color="gray", linestyle="--", linewidth=0.5)
plt.axhline(1, color="gray", linestyle="--", linewidth=0.5)
plt.axvline(0, color="red", linestyle=":", linewidth=1)
plt.title("Exponential Functions")
plt.xlabel("Input ($x$)")
plt.ylabel("Output")
```

```
plt.grid(True)
plt.legend()
plt.show()
```




[69]:
```
print(f"{'x':>6} | {'e^(-x)':>14}")
print("-" * 23)

for value in [-10, 0, 10]:
    print(f"{value:>6} | {np.exp(-value):>14.6g}")
```

```
     x |         e^(-x)
-----------------------
   -10 |        22026.5
     0 |              1
    10 |    4.53999e-05
```

### 1.1 Limits of the Sigmoid

$$\lim_{x\to-\infty}\sigma(x)=\lim_{x\to-\infty}\frac{1}{1+e^{-x}}=\frac{1}{1+\infty}=0$$
$$\lim_{x\to0}\sigma(x)=\lim_{x\to0}\frac{1}{1+e^{-x}}=\frac{1}{1+1}=0.5$$
$$\lim_{x\to\infty}\sigma(x)=\lim_{x\to\infty}\frac{1}{1+e^{-x}}=\frac{1}{1+0}=1$$

The sigmoid function has a **squashing effect**, mapping any input value to a value between 0 and 1. This makes it useful for binary classification problems, where the output can be interpreted as the probability of the input belonging to a certain class.

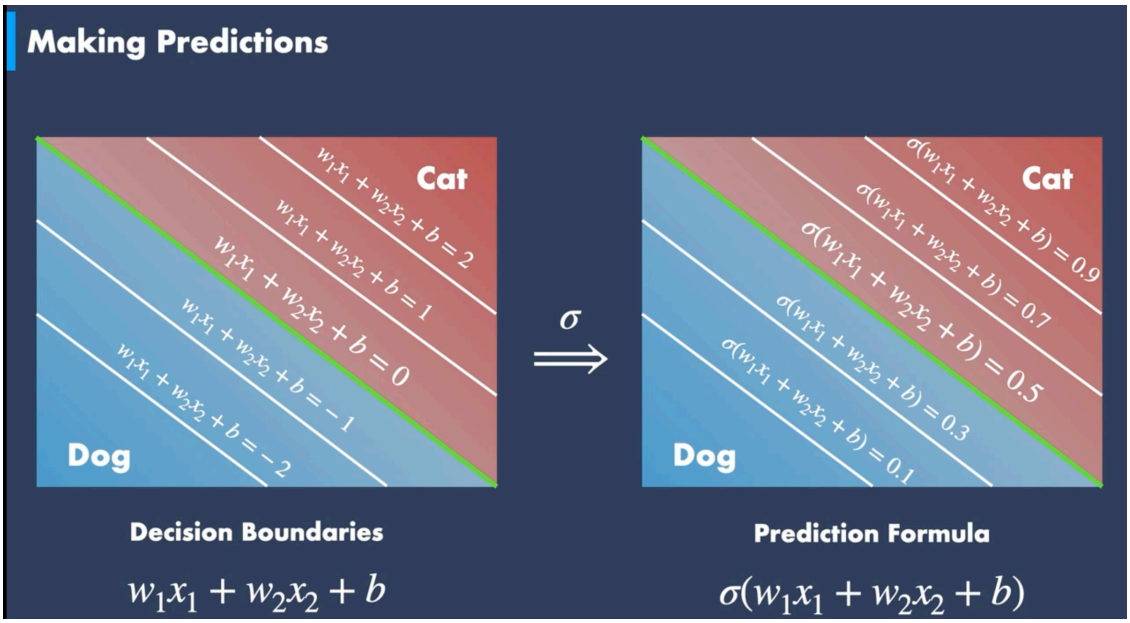
What this means is:

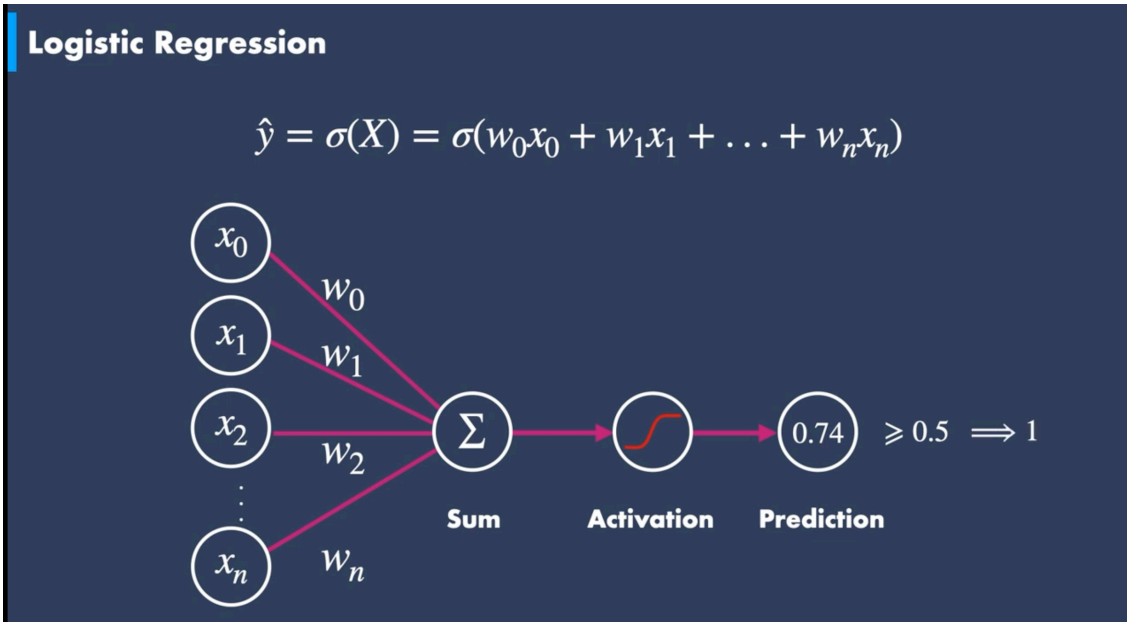
$$-\infty \leq x \leq \infty \Rightarrow 0 < \sigma(x) < 1$$

For any value that x takes, the value of the sigmoid function is always between zero and one.

$$x \in \mathbb{R} \Rightarrow 0 < \sigma(x) < 1$$

## 1.2 Making predictions



If it's more than 0.5 we assign it a value of 1. If it's less, a value of 0.

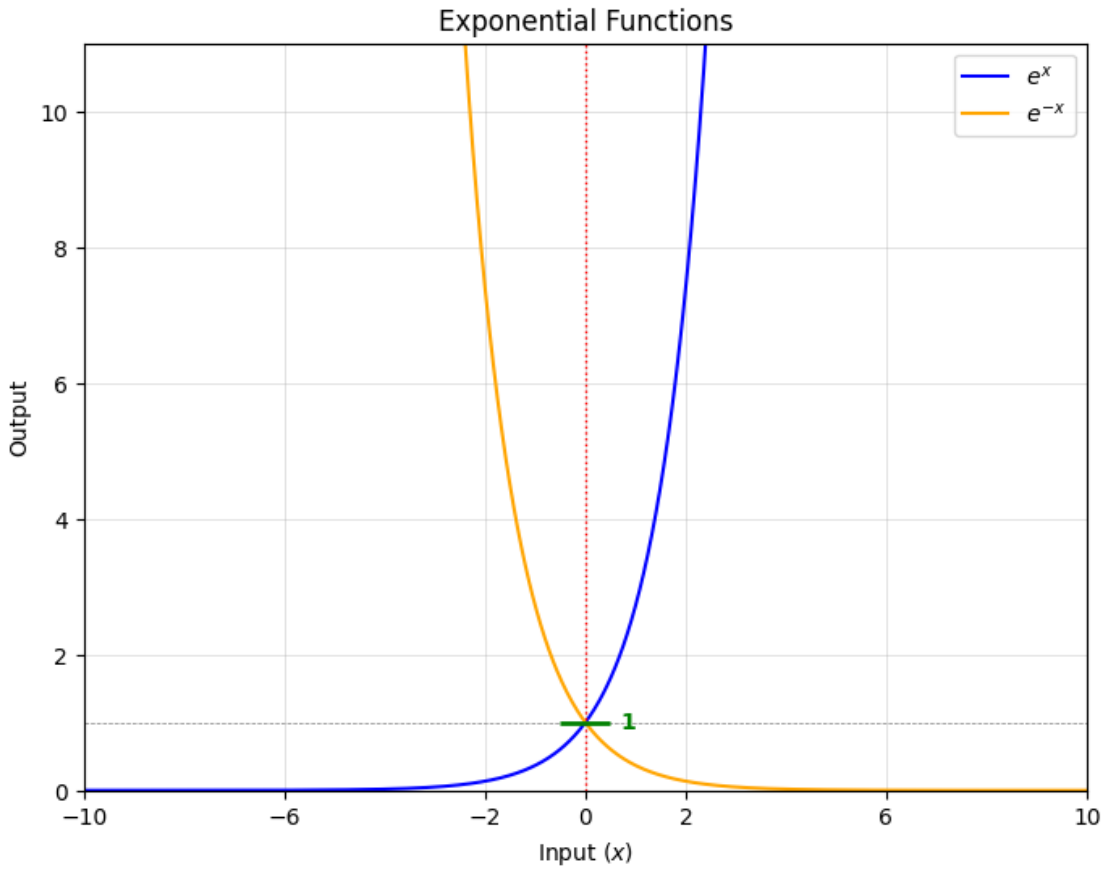
### 1.3 Variations of the graph

In part of the video the $e^x$ and $e^{-x}$ seems to be deformed to make it more visual. Let me play with the functions.

The video uses a wider x-range (−10 to 10) and a limited y-range, which makes the curves look narrower. You can replicate that without changing the functions.

[70]:
```
x = np.linspace(-10, 10, 1001)
y_exp = np.exp(x)
y_exp_negative = np.exp(-x)

# Plot the exponential functions
plt.figure(figsize=(8, 6))
plt.plot(x, y_exp, label=r"$e^x$", color="blue")
plt.plot(x, y_exp_negative, label=r"$e^{-x}$", color="orange")
plt.axhline(0, color="gray", linestyle="--", linewidth=0.5)
plt.axhline(1, color="gray", linestyle="--", linewidth=0.5)
plt.axvline(0, color="red", linestyle=":", linewidth=1)
plt.title("Exponential Functions")
plt.xlabel("Input ($x$)")
plt.ylabel("Output")
plt.grid(True)
plt.legend()
plt.xlim(-10, 10)
plt.ylim(0, 11)
plt.xticks([-10, -6, -2, 0, 2, 6, 10])
plt.hlines(y=1, xmin=-0.5, xmax=0.5, color="green", linewidth=2)
plt.text(0.7, 1, "1", color="green", weight="bold", va="center")
plt.show()
```


Exponential Functions
$e^x$
$e^{-x}$
Output
Input ($x$)
1
10
8
6
4
2
0
−10
−6
−2
0
2
6
10